

PRODUCT REQUIREMENTS DOCUMENT

AI Powered Legal Operations Platform - A bespoke, company-owned, open-source AI platform for the DSN and EqualyzAI legal function, with a path to an external AI plus Law product

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How to read this document

Sections 1 to 4 are the business case and the measurable definition of success. Sections 5 to 8 are the functional specification, written as user stories with acceptance criteria that engineering can estimate and QA can test, together with the end-to-end flows and state model. Sections 9 to 12 are the technical specification: data model, architecture, integrations and interfaces. Sections 13 to 16 cover AI governance, risk tiering, security and the evaluation regime that must pass before any tier is allowed to operate on live matters. Sections 17 and 18 cover delivery: roadmap and risks.

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1. Executive summary

DSN and EqualyzAI intend to offer AI plus Law capability to the market. We cannot credibly sell that capability until we have run it on ourselves. This document specifies the platform that makes the internal legal function measurably faster and better governed, and that doubles as the reference implementation of the product we intend to sell.

1.1 What we are building

A single, company-owned platform that carries a legal request through its whole life: intake, triage, matter creation, drafting from controlled templates and clauses, AI-assisted first drafts, review and issue flagging, approval, execution, archiving, obligation tracking and reporting. Around that spine sit four capabilities that the current toolset does not provide at all: legal inbox intelligence, a searchable institutional memory of negotiation decisions, structured privacy and AI assessment workflows, and live management reporting.

Non-negotiable rule, carried forward from the Legal brief: AI may recommend. An authorised human must confirm. No component of this platform may independently give legal advice, accept risk, approve a contract, sign a document, or send a substantive external communication. Every requirement in this document is subordinate to that rule.

1.2 Why bespoke and open source

Decision	Rationale
Company-owned platform	The legal platform is intended to become a product, not only an internal tool. Owning the experience, the data model and the AI orchestration is the differentiator. A configuration of someone else's suite cannot be licensed to a third party.
Open-source stack	The organisation runs both Microsoft 365 and Google Workspace. Any design that assumes one suite creates a captive dependency and a second-class experience for half the users. An open-source core keeps us portable across identity, mail, storage and model providers.
Provider-neutral AI layer	Model capability, pricing and availability move quickly, and some matters will require a locally hosted model. All model calls go through one gateway so that a route can be changed by configuration, not by code.
Connectors where useful, not as the platform	Self-hosted n8n handles mail collection, notifications, calendar writes and similar plumbing. Business logic, records and legal controls live in the platform, not scattered across automation flows.

Decision	Rationale
Self-hosted deployment	Client matters, counterparty data and privileged material stay inside infrastructure we control, which is also the precondition for the data-sovereignty claim we will make to external customers.

1.3 What good looks like

The target state is not a faster version of today. It is a different operating shape.

Dimension	Today, manual and serial	Target, AI-augmented and parallel
Turnaround	A routine NDA queues behind other work	Minutes, generated from an approved template at intake
Templates	Archived Word versions copied by hand	One versioned clause and template library with review dates
Drafting	Counsel writes and adapts from past versions	AI drafts the first pass, counsel edits, decides and signs
Issue spotting	Deviations found by reading, and by memory	Deviations flagged automatically against the playbook and ranked by severity
Versions	Changes reconstructed after the fact	Tracked differences and an audit trail of who changed what and when
Memory	Precedent lives in individual heads and inboxes	Searchable, source-cited archive of clauses, fallbacks, outcomes and the reasons for them
Visibility	Status rebuilt by hand from several sheets	Live workload, ageing, deadline and delivery dashboards

2. Problem statement and current state

Legal already holds the important material: templates, registers and executed agreements. The failure is not the absence of records, it is the absence of an operating system around them. The Excel workbook is

a serviceable register, but it cannot create work, enforce complete instructions, generate a document, route an approval, read a deadline out of an email, or show management a live view.

2.1 Problems already identified by Legal

Problem	Current effect	Risk created
Reminders and follow-up	Deadlines depend on individuals checking trackers and remembering next actions	Missed filings, renewals and deliverables, and delayed responses
Contract management	Documents, metadata, versions, approvals and obligations are not managed as one lifecycle	Unclear status, inconsistent records, weak post-signature monitoring
Internal legal workflow	Requests arrive through several channels with no standard intake or triage	Incomplete instructions, unclear ownership, avoidable back-and-forth
Form-to-contract drafting	A template is opened, copied and populated by hand before clauses are adapted	Slow first drafts, repeated data entry, accidental use of an outdated template
Inbox triage	Legal manually decides whether an email needs action, awareness or preparation	Urgent work missed, implied legal tasks never become tracked matters
Institutional memory	Negotiation decisions, fallback positions and reasons stay in email and in individual memory	Positions re-litigated, inconsistent concessions, knowledge lost when people move
DPIA and AI assessment	Templates are completed but workflow, evidence, risk flags and sign-off are disconnected	Assessments become late-stage paperwork rather than a control on product release
Visibility and measurement	Priorities, overdue work, turnaround and blockers are assembled manually	Management cannot see load or delivery, and improvement cannot be evidenced

2.2 The underlying issue

DSN-Legal does not need another database. Legal needs a connected operating flow in which a request becomes a matter, a matter produces controlled documents and decisions, and the resulting obligations stay visible long after execution. Every unconnected tool added on top of the current state reproduces the same ambiguity at greater speed.

3. Goals, objectives and non-goals

3.1 Product goals

ID	Goal	Why it matters
LOP-G-01	One controlled front door for all legal work	Removes the multi-channel intake problem and makes instructions complete before work starts
LOP-G-02	One lifecycle record from request to obligation	Ends the split between documents, metadata, approvals and post-signature duties
LOP-G-03	Drafting time reduced by an order of magnitude for standard agreements	Frees counsel time for judgment, which is the scarce resource
LOP-G-04	Nothing falls through the cracks	Deadlines, renewals, filings and promised actions are system-owned, not memory-owned
LOP-G-05	A living institutional memory	Positions, fallbacks and reasons become searchable assets rather than personal knowledge
LOP-G-06	Demonstrable AI governance	Enables safe internal use and becomes a sellable feature of the external product
LOP-G-07	Provider and suite neutrality	Protects against lock-in across identity, mail, storage and model providers

3.2 Design principles

Principle	What it means in practice
Automate the predictable first	If a rule can be stated clearly, implement the rule. Generative AI is reserved for interpretation of unstructured material where it adds genuine value.
Grounded or silent	AI outputs must be grounded in approved templates, clauses, playbooks or the matter record, and must cite their source. Ungrounded generation is a defect, not a feature.
Human confirmation at every substantive step	Suggestions are drafts until a named person accepts them, and the acceptance is recorded.
One source of truth	Only the approved clause library may be presented as house position. Copies are read-only derivatives.
Reversible by design	Every workflow has a manual fallback and an export route, so continuity never depends on one automation.
Separation of the two entities	DSN and EqualyzAI share a data model but not visibility. Entity is a first-class field with enforcement at query level.
Log everything that matters	Prompt, source, model route, user, time, output and final decision are recorded, subject to minimisation and retention rules.
Make the controlled path the easy path	Adoption fails when the portal is slower than an email. Intake must be faster than the habit it replaces.

4.0 Metrics

4.1 Success metrics

ID	Metric	Baseline	Phase 1 target	Phase 3 target
LOP-KPI-01	Median time to acknowledge a legal request	To capture	Under 4 working hours	Under 1 working hour
LOP-KPI-02	Median time from complete instructions to first draft, standard agreement	To capture	Under 60 minutes	Under 10 minutes
LOP-KPI-03	Median end-to-end turnaround, Tier 1 agreements	To capture	Under 1 working day	Under 2 hours
LOP-KPI-04	Median end-to-end turnaround, Tier 2 agreements	To capture	Under 5 working days	Under 3 working days
LOP-KPI-05	Requests arriving with all mandatory fields complete	Low, unmeasured	Over 80 per cent	Over 95 per cent
LOP-KPI-06	Executed contracts with complete metadata and linked obligations	To capture	Over 85 per cent	Over 98 per cent
LOP-KPI-07	Overdue legal tasks and obligations, count and age	To capture	Reduced by 50 per cent	Zero over 14 days overdue
LOP-KPI-08	Time to produce a management update	Hours	Under 15 minutes	On demand, zero manual effort
LOP-KPI-09	Share of legal requests entering through the portal	Near zero	Over 70 per cent	Over 95 per cent
LOP-KPI-10	Counsel hours reclaimed per month	Zero	At least 30 hours	At least 80 hours

4.2 AI quality metrics

These are gates, not dashboards. A capability that falls below its threshold on the golden set is disabled until it passes again. Measurement method is defined in section 16.

ID	Capability	Metric	Gate to enable
LOP-AIQ-01	Inbox classification	Macro F1 across categories	At least 0.90, with recall at least 0.95 for action-required
LOP-AIQ-02	Deadline and date extraction	Precision and recall	Precision at least 0.95, recall at least 0.90
LOP-AIQ-03	Party and entity extraction	Field-level exact match	At least 0.95 on legal name, at least 0.90 on other fields
LOP-AIQ-04	Clause retrieval for drafting	Recall at 5 against the approved library	At least 0.92
LOP-AIQ-05	AI first draft	Unsupported statement rate, human-reviewed	Under 2 per cent of clauses, zero in defined-term and cross-reference checks
LOP-AIQ-06	Redline and deviation flagging	Recall on critical clause deviations	At least 0.95, with false-positive rate under 0.20
LOP-AIQ-07	Obligation extraction from executed contracts	Recall on dated obligations	At least 0.93
LOP-AIQ-08	Institutional memory retrieval	Answer groundedness with citation	At least 0.95 of answers fully attributable to a retrieved source
LOP-AIQ-09	All AI capabilities	Human correction rate, trend	Declining month on month after the first 60 days

5. Users, personas and roles

5.1 Personas

Persona	Needs	Success for them looks like
Requester, business or programme staff	To get an agreement without learning legal process, and to know where it is	One short guided form, a visible status, and a signed document without chasing
Legal counsel	Control of quality and position, with less typing and less chasing	A first draft ready for judgment, deviations already flagged and ranked, precedent one search away
Legal operations	To clear standard work and escalate only genuine exceptions	A queue with clear rules, batch actions and a defensible record of what was cleared
Head of Legal	Load, ageing, risk exposure and delivery evidence	A live dashboard, no manual report assembly, and defensible metrics for management
Executive sponsor	Evidence that the investment produced measurable change	Metrics against baseline, and a credible external product path
External customer	A legal operations capability with sovereignty and control	Isolated tenancy, configurable playbook, exportable data, transparent AI controls

5.2 Roles and permissions model

Access is role-based, entity-scoped and matter-scoped. A user's effective permission is the intersection of role, entity and matter access. Restricted matters override everything and are visible only to explicitly named users.

Role	Can do	Cannot do
Requester	Raise a request, view own matters, upload documents, receive status	See other requesters' matters, see the clause library, see AI traces

Role	Can do	Cannot do
DSN Management	All requester rights plus approve commercial terms assigned to them	Approve legal position, change templates
Legal operations	Triage, clear Tier 1 and Tier 2 standard deviations, generate documents, manage obligations	Approve Tier 3 or Tier 4 positions, amend the clause library, open restricted matters
Counsel	All legal work on assigned matters, approve positions, sign off AI output, propose clause changes	Publish a clause library version alone, change system-wide controls
Head of Legal, or clause owner	Publish clause and template versions, set risk tiers, override escalation, access all non-restricted matters	Alter audit logs, change security configuration
Privacy or DPO role	Run and approve DPIA and AI assessments, view data-processing records	Alter contract positions
AI and Platform administrator	Configure model routes, prompts, skills, evaluation sets, enable or disable capabilities, Manage users, roles, entities, retention, integrations, backups	Read restricted matter content, approve legal output, alter the immutable audit store
Auditor, read only	Read audit trails, evaluation results, decision records	Change any record
External counterparty, limited	Receive a document, view a defined package, sign	Access the platform beyond their own signing session

6. Product scope and module map

The platform is specified as sixteen modules. Phase labels indicate the earliest phase in which the module is delivered to a usable standard, not the phase in which work begins.

ID	Module	Purpose	Phase
M01	Legal portal and guided intake	One controlled front door with conditional, complete request capture	1
M02	Matter management and triage	Matter records, ownership, priority, SLA clock, workload views	1
M03	Template and clause library	Versioned, approved templates, clauses and fallback positions	1 core, 2 full
M04	Document generation	Deterministic assembly of a controlled document from structured facts	1
M05	AI drafting assistant	Grounded first drafts of bespoke agreements from a brief and approved clauses	2
M06	Review, redlining and issue flagging	Compare incoming drafts to the playbook and rank deviations by severity	2
M07	Approval routing and e-signature	Ordered approvals with an audit trail, then controlled execution	1
M08	Executed archive and obligation engine	One authoritative repository and obligations that become tracked tasks	1 core, 2 full
M09	Legal inbox intelligence	Classify, extract and recommend from the shared legal mailbox, never send substantively	2
M10	Chat - Institutional memory and retrieval	Source-cited search across contracts, clauses, decisions and outcomes	2
M11	Privacy, DPIA and AI assessment	Structured assessments connected to products, datasets, vendors and controls	2

ID	Module	Purpose	Phase
M12	Compliance and statutory calendar	Filing calendar, accountable owners, evidence and escalation	2
M13	Counterparty and vendor governance	One counterparty identity linked to contracts, reviews, renewals and performance	2
M14	Reporting and analytics	Live workload, ageing, turnaround, risk and delivery reporting	1 core, 3 full
M15	Administration, access control and audit	Identity, roles, entity separation, retention, immutable audit trail	1

6.1 Module dependency view

Delivery order is constrained by data dependency rather than by preference. The critical path runs from identity and entity separation (M15) through intake and matter records (M01, M02) to the controlled template and clause library (M03), which is the precondition for everything generative. Nothing in M04, M05 or M06 may be built against uncontrolled templates.

Module	Depends on	Note
M01 Intake	M15	Cannot capture entity-scoped requests without identity and entity model
M02 Matter	M01, M15	Matter identifiers are generated at acceptance, not at request
M03 Library	M15	Version ownership requires roles; publication is a controlled act
M04 Generation	M02, M03	Only an approved, published template version may be merged
M05 AI drafting	M03, M04, M10	Grounding requires a published library and working retrieval

Module	Depends on	Note
M06 Flagging	M03	Severity ranking is meaningless without house positions and fallbacks
M07 Approval and signature	M02, M04	Signature binds to an exact document version hash
M08 Archive and obligations	M07	Obligations derive from the executed version, not a draft
M09 Inbox intelligence	M02, M15	Recommendations create matters, so triage must exist first
M10 Memory	M03, M08	Corpus is the library plus executed agreements plus decision records
M11 DPIA	M02, M13	Assessments attach to products, vendors and datasets
M12 Compliance	M02	Filings are recurring obligations on the same task engine as M08
M13 Counterparty	M02	Counterparty identifiers are referenced by every matter and contract
M14 Reporting	M02, M07, M08	Metrics are computed from lifecycle events, not from separate entry
M16 Product layer	All	Only after the internal platform is stable and measured

7. Functional requirements

Each module below states its purpose, its user stories with acceptance criteria, and the business rules that constrain implementation.

7.1 M01, Legal portal and guided intake

The portal is the single controlled front door. Its only real competitor is a direct email, so it must be faster than sending one. Conditional questions ask only what the selected request type requires, and the form cannot be submitted while a mandatory field is missing.

ID	User story	Acceptance criteria	Pri
LOP-M01-US-01	As a requester I want to choose what I need in plain language so I do not have to know legal categories	Landing page lists request types in business language (for example, we want to sign an NDA, we are engaging a consultant, a partner sent us their contract). Each maps to an internal request type. Selection is possible in two clicks or fewer. Free-text fallback exists and routes to triage.	M
LOP-M01-US-02	As a requester I want to be asked only relevant questions so the form is short	Questions render conditionally on request type, entity and value band. No request type shows more than 12 fields on one screen. Progressive disclosure is used for optional detail. Median completion time under 4 minutes measured in user testing.	M
LOP-M01-US-03	As Legal I want submissions to be complete before they reach me	Submission blocked while any mandatory field for the request type is empty. Mandatory set is configurable per request type by the Head of Legal without code change. Attempted submission shows inline errors, never a generic failure.	M
LOP-M01-US-04	As a requester I want to attach supporting documents	Drag and drop and file picker upload, minimum 50 MB per file, accepted types configurable, virus scanned before storage, attached to the request record with uploader and timestamp. Upload failure is retried and reported clearly.	M
LOP-M01-US-05	As a requester I want to declare what data is involved so privacy	Intake asks whether personal data, special-category data, or third-party confidential information is involved, and whether data leaves Nigeria. A yes on	M

ID	User story	Acceptance criteria	Pri
	review is triggered correctly	any of these sets a privacy flag on the matter and notifies the DPO.	
LOP-M01-US-06	As a requester I want to see status without emailing anyone	Requester sees a status timeline for each of their own requests with current stage, owner first name, expected date and last update time. No other requester's data is visible. Status is updated by lifecycle events, not manual entry.	M
LOP-M01-US-07	As Legal I want an acknowledgment to be automatic	On submission the requester receives an acknowledgment within 60 seconds containing the request reference, what happens next and the expected response window. The message is administrative only and contains no legal position.	M
LOP-M01-US-08	As a requester I want to raise a request from where I already work	Requests can be created from a link in Teams and in Google Chat or Gmail, and from a shared portal URL, all authenticating through single sign-on with no second login.	S

Business rules, M01

- Information-only correspondence must not create a matter. A request remains a request until Legal accepts it.
- A preliminary enquiry may be answered and closed without a matter number.
- The requester never selects the risk tier. Tier is derived by rule and may be overridden only by Legal.

7.2 M02, Matter management and triage

A matter is the container for a piece of legal work. It carries ownership, priority, deadline, next action and status, and it is the object that every document, communication, approval and obligation attaches to.

ID	User story	Acceptance criteria	Pri
LOP-M02-US-01	As Legal operations I want a triage queue so nothing sits unassigned	New requests appear in a queue sorted by declared deadline and derived urgency. Queue shows entity, request type, counterparty, privacy flag, age and suggested tier. Accepting a request generates a matter number and assigns an owner.	M
LOP-M02-US-02	As Legal I want the system to propose a tier and a lawyer	On acceptance the system proposes a risk tier from configurable rules on request type, value band, counterparty class, data sensitivity and template deviation, and proposes an owner from workload and specialism. Both proposals are editable, and any change is recorded with a reason.	M
LOP-M02-US-03	As counsel I want one view of my open work	My matters view lists matter, entity, counterparty, type, tier, status, next action, due date, days open and blocker. Sortable and filterable. Loads in under 3 seconds at the 95th percentile with 5,000 matters in the system.	M
LOP-M02-US-04	As the Head of Legal I want a service clock on every matter	Each matter type has a configurable SLA. The clock starts at acceptance, pauses on a recorded wait-on-requester or wait-on-counterparty state, and resumes automatically. Breach and near-breach are flagged and reportable.	M
LOP-M02-US-05	As counsel I want to record a decision and its reason so the memory survives me	Every matter has a decision log. Entries capture decision, reason, clause reference, alternatives considered, who decided and when. Entries are indexed into institutional memory (M10) and cannot be silently deleted, only superseded.	M
LOP-M02-US-06	As Legal I want related work grouped, not duplicated	A matter can be linked to a parent matter, to other matters on the same counterparty, and to amendments of a prior contract. Correspondence on existing work links to the existing matter rather than creating a new one.	M

ID	User story	Acceptance criteria	Pri
LOP-M02-US-07	As the Head of Legal I want to reassign work without losing history	Reassignment records previous and new owner, reason and timestamp, notifies both, and retains the full history on the matter.	S
LOP-M02-US-08	As Legal I want restricted matters to be genuinely restricted	A matter can be marked restricted by counsel or above. Restricted matters are excluded from all lists, search, retrieval indexes, dashboards and exports for users not explicitly named, and access attempts are logged.	M

7.3 M03, Template and clause library

This module is the single source of truth for house position. Everything generative depends on it, so it is built before AI drafting rather than alongside it.

ID	User story	Acceptance criteria	Pri
LOP-M03-US-01	As the clause owner I want templates to be versioned and approved	Templates have owner, status (draft, approved, superseded, withdrawn), version number, approval date, approver, effective date and review date. Only an approved, effective version can be used for generation. Superseded versions remain readable and are never deleted.	M
LOP-M03-US-02	As the clause owner I want a clause library with fallback positions	Each clause has a category, house position text, up to three ranked fallback positions, an unacceptable-position description, usage conditions, risk notes and an owner. Fallbacks carry the authority level required to concede them.	M
LOP-M03-US-03	As counsel I want to propose a clause change through review, not by editing in place	Proposals create a new draft version with a differences view against the current version, route to the clause owner for approval, and publish atomically. Concurrent proposals are detected and must be merged before publication.	M

ID	User story	Acceptance criteria	Pri
LOP-M03-US-04	As Legal I want templates to declare their variables	Each template declares its merge variables with type, validation rule, source field in the data model and whether it is mandatory. Generation fails safely and reports the missing variable rather than emitting a placeholder into a document.	M
LOP-M03-US-05	As the Head of Legal I want to know when a template is going stale	Each template and clause has a review date. Owners are notified 30 days before, and overdue reviews appear on the Legal dashboard and in the monthly report.	S
LOP-M03-US-06	As Legal I want templates scoped to the right entity and jurisdiction	Templates and clauses carry entity applicability (DSN, EqualyzaI or both), jurisdiction and language. Generation offers only applicable versions.	M
LOP-M03-US-07	As Legal I want to import our existing templates without losing what makes them ours	Import accepts existing Word files, extracts candidate clauses, proposes a clause breakdown for human confirmation, and records provenance. No imported clause becomes approved without explicit human approval.	S

Hard constraint

Only the approved clause library may be presented anywhere in the platform as house position. AI-generated clause text that is not traceable to an approved clause or an approved fallback must be visibly marked as novel and unapproved wherever it appears, including in exported documents and in redline suggestions.

7.4 M04, Document generation

Generation is deterministic. Given the same structured facts and the same template version, the output must be byte-identical. Nothing generative happens here, which is what makes Tier 1 auto-issue defensible.

ID	User story	Acceptance criteria	Pri
LOP-M04-US-01	As Legal operations I want a document assembled from the matter record	Selecting an approved template on a matter merges party details, dates, values, scope and deliverables from structured fields, and produces a .docx and a PDF preview. No manual retyping of any field already held in the record.	M
LOP-M04-US-02	As Legal I want conditional content, not twenty near-identical templates	Templates support conditional sections and repeating blocks driven by matter fields, for example a schedule of deliverables or a data-processing annex included only when the privacy flag is set.	M
LOP-M04-US-03	As Legal I want generation to be reproducible and attributable	Every generated document records template ID and version, clause versions used, the exact input values, the generating user, the timestamp and a content hash. Regeneration with the same inputs produces an identical file.	M
LOP-M04-US-04	As Legal I want Tier 1 documents issued without a drafting cycle	For request types configured as Tier 1 and with no deviation from the approved template, the platform generates, routes for signature and files the agreement automatically, and adds it to the monthly quality sample.	M
LOP-M04-US-05	As a requester I want the document in a format I can actually use	Output available as .docx and PDF. Documents can be opened in Word or Google Docs through the relevant integration without breaking the version link back to the platform record.	M
LOP-M04-US-06	As Legal I want generation blocked when the record is not ready	Generation is refused with a clear reason when a mandatory variable is missing, the template version is not approved or effective, the counterparty record is incomplete, or a required approval is outstanding.	M

7.5 M05, AI drafting assistant

For bespoke agreements a deterministic merge is not enough. The assistant produces a first pass from a brief plus the approved clause library, and it must show its work. This is the module where discipline matters most.

ID	User story	Acceptance criteria	Pri
LOP-M05-US-01	As counsel I want a first draft from a short brief	Given a matter, a request type and a brief, the assistant returns a structured draft agreement built from approved clauses where they exist, within 90 seconds at the 95th percentile. Draft opens in the platform editor with clause-level provenance visible.	M
LOP-M05-US-02	As counsel I want to see where every clause came from	Each clause in the draft shows its source: approved clause and version, approved fallback, prior executed agreement (with matter reference), or novel text. Novel text is highlighted distinctly and counted in a header summary.	M
LOP-M05-US-03	As counsel I want house style enforced, not suggested	Drafting follows a configurable house style set: defined-term usage, numbering, cross-reference format, party naming, currency and date conventions, governing-law phrasing. Violations are corrected before the draft is presented and are listed in a style report.	M
LOP-M05-US-04	As counsel I want the assistant to tell me what it could not do	The draft includes an explicit open-items list: missing facts, decisions requiring instruction, clauses with no approved position, and assumptions made. An assumption not surfaced in this list is a defect.	M
LOP-M05-US-05	As counsel I want internal consistency checked mechanically	Before presentation, deterministic checks run on defined terms used but not defined, terms defined but unused, broken cross-references, duplicated numbering, inconsistent party names, date logic and blank placeholders. All failures are reported, and the assistant may not present a draft with an unreported failure.	M
LOP-M05-US-06	As counsel I want to iterate in place	Counsel can instruct changes in natural language on a selected clause, see the proposed replacement as a difference, and accept, reject or edit. Every accepted change is attributed to the counsel, not to the model.	M

ID	User story	Acceptance criteria	Pri
LOP-M05-US-07	As the AI administrator I want drafting disabled per agreement type	Drafting availability is configurable per request type and tier, and can be disabled instantly for a type or globally without deployment.	M
LOP-M05-US-08	As counsel I want the draft to remember this counterparty	Where prior agreements with the same counterparty exist, the assistant surfaces previously agreed positions and prior concessions with matter references, and warns when the current draft contradicts a prior position.	S

7.6 M06, Review, redlining and issue flagging

Most incoming risk arrives as the counterparty's paper. This module compares what arrived against what we accept, and ranks the difference so attention goes to what matters.

ID	User story	Acceptance criteria	Pri
LOP-M06-US-01	As counsel I want an incoming draft compared to our playbook	Uploading a counterparty draft produces a clause-by-clause comparison against the applicable playbook, with a severity for each finding: critical, material, minor or acceptable. Findings link to the affected text and to the house position.	M
LOP-M06-US-02	As counsel I want missing protections flagged, not only altered ones	The comparison reports required clauses that are absent altogether, including limitation of liability, confidentiality, data protection, IP ownership, termination, governing law and dispute resolution, according to the playbook for that agreement type.	M
LOP-M06-US-03	As counsel I want a suggested response per finding	Each finding offers the house position, the ranked acceptable fallbacks, the authority level needed to concede, and suggested redline text. Suggestions are drafts and are marked as such until accepted.	M
LOP-M06-US-04	As counsel I want to produce a real redline	Accepted suggestions are written back as tracked changes in a .docx that opens correctly in Word and in Google Docs, attributed to the platform on behalf	M

ID	User story	Acceptance criteria	Pri
		of the named counsel, with a comment carrying the reason where configured.	
LOP-M06-US-05	As Legal operations I want to clear standard deviations without waiting for counsel	For Tier 2 matters, findings classified as minor and matching a pre-approved fallback can be cleared by Legal operations. Every clearance records the rule applied, the person and the time. Anything else escalates automatically.	M
LOP-M06-US-06	As the Head of Legal I want deviation patterns visible	Reporting shows which clauses are most frequently challenged, by which counterparty class, with what outcome, so the playbook can be revised on evidence rather than impression.	S

7.7 M07, Approval routing and e-signature

ID	User story	Acceptance criteria	Pri
LOP-M07-US-01	As the Head of Legal I want approvals routed in the correct order	Approval chains are configurable by entity, agreement type, value band and risk tier, and support sequential, parallel and conditional steps covering Finance, programme, executive and signatory approval. The chain applied is recorded on the matter.	M
LOP-M07-US-02	As an approver I want to approve without logging into another system	Approvers receive a notification in email, Teams or Google Chat with the document, the summary of terms, the deviations accepted and a secure action link. Action requires authenticated single sign-on. Approval, rejection and comments are recorded with time.	M
LOP-M07-US-03	As Legal I want approvals to bind to an exact version	Approval attaches to a document content hash. Any subsequent edit invalidates outstanding approvals, notifies affected approvers and requires re-approval. A signature request cannot be issued for an unapproved hash.	M

ID	User story	Acceptance criteria	Pri
LOP-M07-US-04	As Legal I want overdue approvals to escalate on their own	Reminders and escalation follow a configurable schedule to the approver, then their delegate, then the escalation owner. Escalation events are visible on the matter and in reporting.	M
LOP-M07-US-05	As Legal I want electronic execution inside our own stack	Signature is requested through the self-hosted e-signature service, supports ordered multi-party signing, records signer identity, IP, timestamp and audit certificate, and returns the executed file to the platform automatically on completion.	M
LOP-M07-US-06	As Legal I want a manual fallback for wet-ink execution	A matter can be marked as executed outside the platform, requiring upload of the signed copy, signature date, signatories and a reason. Metadata completeness rules still apply.	M
LOP-M07-US-07	As Legal I want to withdraw a signature request	An outstanding signature request can be cancelled by counsel or above, with a reason, and the counterparty link is invalidated immediately.	S

7.8 M08, Executed archive and obligation engine

The archive is the authoritative record, and the obligation engine is what makes post-signature life visible. This is where the reminders and follow-up problem is actually solved.

ID	User story	Acceptance criteria	Pri
LOP-M08-US-01	As Legal I want one authoritative executed copy	On completion the executed PDF, all approvals, the signature certificate and full metadata are stored as an immutable record with a content hash. The record is marked authoritative and any later upload is a linked amendment, not a replacement.	M
LOP-M08-US-02	As Legal I want obligations extracted, then confirmed	On archiving, the platform proposes obligations covering deliverables, payment milestones, reporting duties, renewal dates, notice periods, termination windows and conditions precedent. Each proposal shows the clause it came from. Legal	M

ID	User story	Acceptance criteria	Pri
		confirms, edits or rejects, and confirmation creates tracked tasks.	
LOP-M08-US-03	As an obligation owner I want to be reminded in time to act	Each obligation has an owner, due date, recurrence, lead time and escalation rule. Reminders reach the owner at configurable lead times, escalate on breach and appear in the owner's working environment, not only in the platform.	M
LOP-M08-US-04	As Legal I want renewals to stop being surprises	Renewal and notice windows generate a task at the notice deadline minus a configurable lead time, defaulting to 60 days, with the decision options recorded: renew, renegotiate, terminate or allow to lapse.	M
LOP-M08-US-05	As an obligation owner I want to record evidence of completion	Completion requires a status and, where configured, evidence such as a document, link or note. Completion is timestamped and attributable, and evidence is retained with the obligation.	M
LOP-M08-US-06	As the Head of Legal I want a single obligation calendar	A calendar view across entities, filterable by owner, counterparty, type and due window, exportable, and available as a subscribable feed into both Microsoft and Google calendars.	M
LOP-M08-US-07	As Legal I want to find any executed agreement in seconds	Search across counterparty, agreement type, entity, value, date range, clause content and obligation status returns results in under 2 seconds at the 95th percentile, respecting restricted-matter rules.	M

7.9 M09, Legal inbox intelligence

The shared legal inbox carries three different things at once: direct instructions, awareness-only messages, and conversations that imply future legal work without assigning it. The third category is where work is currently lost. The platform helps Legal see the difference and never speaks for Legal.

ID	User story	Acceptance criteria	Pri
LOP-M09-US-01	As Legal I want the approved mailbox monitored, and only that mailbox	The platform connects to explicitly named mailboxes only, through Microsoft Graph or Gmail API with least-privilege scopes. The mailbox list is configuration, is displayed in the admin interface, and any change is an audited event. Personal mailboxes and broad archives are never ingested.	M
LOP-M09-US-02	As Legal I want each message classified	Every new message receives a suggested classification from a fixed set: action required, deadline present, awareness only, possible contract, privacy issue, vendor issue, unclear. Confidence is shown. Legal can correct the classification, and corrections feed the evaluation set.	M
LOP-M09-US-03	As Legal I want the facts pulled out for me	Extraction identifies named parties, dates and deadlines, requested deliverables, referenced documents, monetary values and the apparent requester. All extracted values are suggestions until confirmed, and each shows the sentence it came from.	M
LOP-M09-US-04	As Legal I want a recommended next step, not an action taken	The platform proposes an acknowledgment draft, a matter type, a priority and a responsible person. Nothing is sent and no matter is created until Legal confirms. Acknowledgment drafts are administrative and contain no legal position, advice or commitment.	M
LOP-M09-US-05	As Legal I want implied work surfaced before it becomes urgent	Messages that imply future legal work without assigning it are flagged with the phrase that triggered the flag, and are listed in a watch view with an ageing clock, separate from the action queue.	M
LOP-M09-US-06	As Legal I want deadlines and silence both escalated	The platform flags approaching extracted deadlines, high-risk language, and messages awaiting a response beyond a configurable period. Escalation rules and recipients are set by Legal.	M

ID	User story	Acceptance criteria	Pri
LOP-M09-US-07	As the AI administrator I want the accuracy of this module measured continuously	False positives, false negatives, correction rate and time saved are recorded per category and reported monthly. The module cannot be expanded to new mailboxes or categories until its gate in section 4.2 is met.	M

Worked example, carried from the Legal brief

An email says, please conclude the partnership before 18 August. The platform detects the date, suggests a partnership-contract matter, proposes an administrative acknowledgment and lists the information that appears to be missing. Legal confirms the matter and the response. Nothing substantive leaves the organisation automatically.

7.10 M10, Chat - Institutional memory and retrieval

This is the module that converts a legal function into an institution. It answers not only what we agreed, but why we agreed it and what we refused.

ID	User story	Acceptance criteria	Pri
LOP-M10-US-01	As counsel I want to ask a question and get a cited answer	Natural-language questions over the approved library, executed agreements, decision logs, negotiation outcomes and playbooks return an answer with inline citations to the specific clause, matter or decision entry. Any statement without a citation is suppressed rather than shown.	M
LOP-M10-US-02	As counsel I want to know our position history on a clause	For any clause category the platform shows the current house position, the fallbacks, every recorded deviation with counterparty and outcome, and the authority who approved each concession.	M
LOP-M10-US-03	As counsel I want counterparty history in one place	A counterparty view shows all matters, contracts, positions taken, concessions granted, disputes, obligations outstanding and the negotiation behaviour observed, in date order.	M

ID	User story	Acceptance criteria	Pri
LOP-M10-US-04	As Legal I want retrieval to respect permissions absolutely	Retrieval is filtered by entity, role and matter access before ranking, not after. Restricted matters are excluded from the index for unauthorised users, and no snippet, title or citation may leak from a record the user cannot open.	M
LOP-M10-US-05	As Legal I want the memory to build itself from normal work	Decision log entries, accepted redlines, approved fallback usage, and executed agreements are indexed automatically on the events that create them. No separate knowledge-entry task exists.	M
LOP-M10-US-06	As the Head of Legal I want stale or superseded knowledge marked	Superseded clause versions and withdrawn positions remain retrievable but are labelled clearly, and the current position is always shown first.	S

7.11 M11, Privacy, DPIA and AI assessment

For a data and AI organisation, privacy and responsible-AI review belongs inside product development, not at the end of it as a form. This module makes assessment a workflow with evidence and an owner.

ID	User story	Acceptance criteria	Pri
LOP-M11-US-01	As the DPO I want assessments as structured workflows	DPIA and AI assessment are structured records with stages, owners, due dates and status, linked to a product, dataset, vendor or matter. Stages route between Product, Engineering, Legal and the accountable business owner.	M
LOP-M11-US-02	As the DPO I want the assessment to capture what actually matters	Each assessment captures purpose, intended users, affected persons, business owner, data categories and sources, legal basis, retention, hosting locations and transfers, models, vendors, subprocessors, connectors, datasets, material contractual terms, potential harms, bias, security threats, performance limits, human oversight, controls, testing evidence, approval conditions, review date and residual-risk owner.	M

ID	User story	Acceptance criteria	Pri
LOP-M11-US-03	As the accountable business owner I want to accept, mitigate or escalate residual risk explicitly	Closure requires a recorded residual-risk decision by the named accountable owner, with reason. The platform will not close an assessment with an unassigned residual risk.	M
LOP-M11-US-04	As Engineering I want to know what was approved and on what condition	Approved assessments produce a machine-readable record of conditions and controls, and outstanding conditions generate tracked tasks on the same engine as M08.	M
LOP-M11-US-05	As the DPO I want reassessment triggered by change	Material change to purpose, data categories, model, vendor or transfer route triggers a reassessment task. Review dates generate reminders in advance.	M
LOP-M11-US-06	As Legal I want assessments linked to the contracts that govern them	An assessment links to the vendor contract, the data-processing terms and the relevant clauses, so a change of terms is visible from the assessment and the reverse.	S

7.12 M12, Compliance and statutory calendar

ID	User story	Acceptance criteria	Pri
LOP-M12-US-01	As Legal I want a filing calendar with owners	Compliance items record requirement, entity, jurisdiction, filing date, recurrence, accountable owner, evidence requirement and next due date, and generate tasks with lead times and escalation.	M
LOP-M12-US-02	As Legal I want evidence of filing retained with the item	Completion requires upload of filing evidence where configured, retained with date, reference number and the person who filed.	M
LOP-M12-US-03	As the Head of Legal I want a compliance status view per entity	A view shows, per entity, items due, overdue, completed this period and evidence gaps, exportable for board or funder reporting.	M

ID	User story	Acceptance criteria	Pri
LOP-M12-US-04	As Legal I want statutory changes handled as a controlled update	Changing a requirement creates a version of the compliance item with effective date, so historical compliance can be assessed against the rule that applied at the time.	S

7.13 M13, Counterparty and vendor governance

ID	User story	Acceptance criteria	Pri
LOP-M13-US-01	As Legal I want one permanent identity per counterparty	Counterparties have a system-generated permanent identifier, legal name, trading names, type, registration number, jurisdiction, addresses and contacts. Name changes update the record without changing the identifier, and history is retained.	M
LOP-M13-US-02	As Legal I want duplicate counterparties prevented at entry	Creation performs fuzzy matching on name, registration number and domain, warns on likely duplicates and offers merge. Merges retain both prior identifiers as aliases.	M
LOP-M13-US-03	As a service owner I want vendor governance in one record	Vendor records link contracts, security reviews, privacy and data-processing terms, subprocessors, renewal dates, service owner, spend band and performance or incident notes.	M
LOP-M13-US-04	As the DPO I want vendor risk visible before renewal	The renewal task for a vendor contract surfaces outstanding security findings, expired assessments and unresolved performance issues.	S

7.14 M14, Reporting and analytics

ID	User story	Acceptance criteria	Pri
LOP-M14-US-01	As the Head of Legal I want a live operational dashboard	Dashboard shows open matters by owner, tier and entity, ageing buckets, SLA breaches, matters blocked and awaiting whom, and turnaround trend. Data is	M

ID	User story	Acceptance criteria	Pri
		computed from lifecycle events, refreshed at least every 15 minutes.	
LOP-M14-US-02	As the executive sponsor I want the improvement metrics on one page	A metrics view reports every KPI in section 4.1 against baseline and target, with the measurement definition visible and a monthly export.	M
LOP-M14-US-03	As the Head of Legal I want risk and exposure reporting	Reporting shows deviations accepted by severity and authority, clauses most frequently conceded, contracts with unusual liability positions, and obligations at risk.	S
LOP-M14-US-04	As the AI and Platform administrator I want an AI usage and quality report	Report shows requests by capability, model route, token and cost, latency, human correction rate, evaluation results against gates, and every capability currently disabled with the reason.	M
LOP-M14-US-05	As management I want the weekly update generated, not written	The platform generates a weekly legal automation progress update covering delivery, volumes, turnaround, blockers and next actions, ready for the weekly tracker with no manual assembly.	M
LOP-M14-US-06	As Legal I want reports respecting entity separation	All reporting is entity-scoped by default. Cross-entity views require an explicit permission and are logged.	M

7.15 M15, Administration, access control and audit

ID	User story	Acceptance criteria	Pri
LOP-M15-US-02	As IT I want multi-factor authentication enforced	MFA is enforced at the identity provider for all users and re-authentication is required for signature, clause publication, restricted-matter access and administrative changes.	M

ID	User story	Acceptance criteria	Pri
LOP-M15-US-03	As Legal I want an immutable audit trail	Every create, read of restricted content, update, delete, approval, generation, AI call, permission change and export is recorded as an append-only event with actor, time, entity, object, before and after state where applicable. The store is write-once for the retention period, and administrators cannot alter it.	M
LOP-M15-US-04	As the platform administrator I want retention and deletion to be policy, not habit	Retention schedules are configurable per record class, with legal hold, defensible deletion, deletion approval and a deletion certificate. Records under hold cannot be deleted by any role.	M
LOP-M15-US-05	As IT security I want data-loss controls at the boundary	Export, download, share-link creation and connector egress are permission-controlled, logged, rate-limited and subject to configurable rules on classification. Bulk export requires approval by a second authorised user.	M
LOP-M15-US-06	As the platform administrator I want configuration without deployment	Request types, mandatory fields, tier rules, approval chains, SLA targets, reminder schedules, playbooks, prompts, model routes and retention schedules are configurable in the administration interface, versioned, and changes are audited.	M
LOP-M15-US-07	As IT I want a working disaster recovery position	Automated encrypted backups of database, object store and audit store, with restore tested at least quarterly against the recovery objectives in section 15.	M

8. End-to-end flows and state model

8.1 The target lifecycle

Six stages, one record. Read left to right.

Stage	What happens	Who acts	System behaviour
1. Request	A new request comes in via email or a business user completes a guided form for the type of support needed	Requester	Validates completeness, stores attachments, acknowledges within 60 seconds, creates a request record
2. Triage	The request is accepted, or answered and closed, or returned for information	Legal operations, counsel	Proposes tier and owner, generates the matter identifier on acceptance, starts the SLA clock
3. Draft and review	Approved templates are populated, or a bespoke draft is produced and reviewed	Legal operations, counsel	Deterministic merge or grounded AI draft, consistency checks, deviation flagging with severity
4. Approve and sign	The matter follows the applicable approval chain, then is executed	Approvers, signatories	Binds approvals to a content hash, invalidates on edit, requests signature, returns the executed file
5. Store and extract	The executed copy, approvals and metadata are preserved	Legal operations	Immutable archive record, hash, proposed obligations for human confirmation
6. Track and report	Obligations, renewals, filings and metrics run from the record	Obligation owners, Head of Legal	Tasks, reminders, escalation, dashboards, weekly update generation

8.2 Matter state model

State	Meaning	Permitted next states
Submitted	Request received, not yet triaged	In triage, Returned for information, Closed without matter

State	Meaning	Permitted next states
Returned for information	Instructions incomplete, awaiting requester	In triage, Closed without matter
In triage	Being assessed for tier, owner and acceptance	Accepted, Returned for information, Closed without matter
Accepted	Matter created, owner assigned, SLA running	Drafting, Awaiting counterparty, On hold
Drafting	Document being generated, drafted or reviewed	In review, Awaiting counterparty, On hold
In review	Legal review or deviation clearance underway	In approval, Drafting, Escalated
Escalated	Outside playbook, awaiting named counsel or authority	In review, In approval, On hold
In approval	Approval chain running	Awaiting signature, In review (on rejection or edit)
Awaiting signature	Signature requested, not yet complete	Executed, In approval (on withdrawal or edit)
Executed	Fully signed and archived	Active
Active	Live agreement with obligations being tracked	Amended, Expired, Terminated
Amended	Amendment executed, linked to the original	Active
Expired or Terminated	No longer live, obligations closed or carried	Archived
On hold	Paused with reason, SLA clock paused	Previous state, Closed

State	Meaning	Permitted next states
Closed without matter	Enquiry answered, no matter created	Terminal

Every transition records actor, timestamp, reason where the transition is a reversal or a hold, and any invalidated approvals. Transitions are the sole source for turnaround metrics, so no metric is computed from manually entered dates.

8.3 Identifier scheme

Identifiers are system-generated, permanent and never reused. A matter identifier must not depend on a counterparty name, because names and spellings change while identity should not.

Identifier	Purpose	Format	Example
Counterparty ID	Permanent identity for an external organisation or individual	CPT-<sequence>	CPT-0047
Matter number	Container for a piece of legal work	<entity>-<practice>-<year>-<sequence>	EAI-COM-2026-0011
Contract ID	Identity for a proposed or executed agreement	<entity>-CON-<year>-<sequence>	EAI-CON-2026-0038
Request reference	Pre-matter reference given to the requester at intake	REQ-<year>-<sequence>	REQ-2026-01184
Template version	Approved template and its version	TPL-<type>-v<major>.<minor>	TPL-NDA-v3.1
Clause version	Approved clause and its version	CLS-<category>-v<major>.<minor>	CLS-LIAB-v2.0
Obligation ID	A tracked duty arising from a contract	OBL-<contract sequence>-<sequence>	OBL-0038-04
Assessment ID	DPIA or AI assessment record	ASM-<year>-<sequence>	ASM-2026-0022

Identifier rules

- Entity prefixes: DSN for Data Science Nigeria, EAI for EqualyzAI. Practice codes are configurable, for example COM for commercial, EMP for employment, IPR for intellectual property, DPR for data protection, CRP for corporate.
- One counterparty may have many matters. One matter may contain several agreements, amendments, communications, assessments and obligations.
- Information-only correspondence does not generate a matter number.
- An actionable request generates a matter number at acceptance, not at submission.
- Correspondence about existing work links to the existing matter rather than creating a new one.

9. Data model

A single data model serves both entities. Entity is a mandatory attribute on every record, and permission enforcement is applied at query level rather than in the presentation layer. The model below is the logical specification. Physical implementation is PostgreSQL with row-level security.

9.1 Core entities

Record	Key attributes and relationships
Organisation	Entity code (DSN, EAI), legal name, registration details, default jurisdiction, branding, retention profile. Parent of every other record.
User	Identity provider subject, name, work email, entity memberships, roles, specialisms, delegate, active status, last login. Provisioned from Entra ID or Google Workspace.
Request	Request reference, entity, requester, request type, subject, purpose, counterparty (proposed), required date, value band, data-involvement flags, attachments, status, triage notes.
Matter	Matter number, entity, request link, request type, practice area, requester, business owner, responsible lawyer, priority, risk tier, status, next action, due date, SLA target, SLA clock state, restricted flag, linked documents, communications, decisions.
Contract	Contract ID, matter, counterparty, agreement type, entity, effective date, term, renewal type, notice period, value and currency, governing law, version, approvals, signature status, executed copy reference, content hash, obligations, amendment chain.
Document	Document ID, matter or contract, type (draft, redline, executed, evidence, correspondence), template and clause versions used, generated-by, content hash, storage reference, classification, version chain, immutability flag.
Clause	Clause ID, category, version, status, owner, house position text, ranked fallbacks with required authority, unacceptable positions, usage conditions, entity applicability, jurisdiction, risk notes, review date.
Template	Template ID, agreement type, version, status, owner, approval date, effective date, review date, declared variables with validation, conditional sections, clause references, entity applicability, jurisdiction, language.

Record	Key attributes and relationships
Task or obligation	Obligation ID, matter or contract, source clause reference, description, type, owner, due date, recurrence, lead time, escalation rule, evidence requirement, status, completion evidence, completed-by and when.
Counterparty	Counterparty ID, legal name, aliases, type, registration number, jurisdiction, addresses, contacts, relationship class, risk class, linked matters and contracts, merge history.
Vendor	Counterparty link, service owner, contracts, security review status and date, privacy and data-processing terms, subprocessors, hosting locations, renewal date, spend band, performance and incident notes.
Product or dataset	Name, purpose, owner, users, datasets, models, vendors, linked DPIA and AI assessment, controls, approval status, review date.
Assessment	Assessment ID, type (DPIA, AI assessment), subject link, stage, owners per stage, captured fields, identified risks, controls, testing evidence, conditions, residual-risk decision and owner, approval, review date.
Compliance item	Requirement, entity, jurisdiction, filing date, recurrence, accountable owner, evidence requirement, evidence, next due date, version and effective date.
Communication	Source mailbox, direction, participants, subject, received or sent time, classification and confidence, extracted values with source sentence, linked matter, actions taken, human confirmations.
Decision record	Matter or clause link, decision, reason, alternatives considered, clause references, authority level, decided-by, decided-at, superseded-by.
Approval	Matter, contract, document hash, chain definition applied, step, approver, delegate used, decision, comments, decided-at, invalidated-by event.
AI interaction	Capability, matter link, user, prompt reference, retrieved source references, model route and version, parameters, output reference, tokens, cost, latency, human decision (accepted, edited, rejected), correction detail.
Audit event	Actor, timestamp, entity, object type and ID, action, before and after state where applicable, IP, session, result. Append-only.

Record	Key attributes and relationships
Tenant (phase 4)	Tenant ID, customer, hosting region, encryption key reference, permitted providers and model routes, entitlement limits, configuration set, metering.

9.2 Data classification

Classification is a mandatory attribute on documents, communications and matters, and it governs which model routes and which connectors may be used. See section 13.4.

Class	Definition	Examples	AI handling
Public	Already published or intended for publication	Published policies, marketing terms	Any approved route
Internal	Ordinary business information, not for external release	Internal templates, process notes	Approved enterprise route with no training use
Confidential	Commercially or legally sensitive, harmful if disclosed	Draft agreements, negotiation positions, counterparty terms	Approved enterprise route, no training use, logging with minimisation, human review mandatory
Restricted	Privileged, special-category personal data, credentials, or matters under special secrecy	Privileged investigations, employee health data, security keys	No general-purpose model. Documented exception plus a specially approved environment, or locally hosted model only

10. Architecture and technology stack

The platform is a modular monolith at the application layer with clearly separated service boundaries, deployed as containers on infrastructure we control. A modular monolith is chosen deliberately over microservices for phase 1 to 3: the team is small, the domain is transactional, and premature service decomposition would cost more in operations than it returns.

10.1 Layered view

Layer	Responsibility	Selected technology	Why
Presentation	Portal, legal workspace, admin, dashboards	Next.js with React and TypeScript, Tailwind CSS, component library	Mature, hireable, server rendering for speed, one codebase for portal and workspace
Application and API	Business logic, lifecycle, permissions, orchestration	Python with FastAPI, Pydantic, SQLAlchemy, Alembic	Same language as the AI layer, strong typing at the boundary, fast to build and test
Records	Transactional store and vector store	PostgreSQL 18 with pgvector and row-level security	One database for records and embeddings reduces operational surface and keeps permission filtering in one place
Search	Keyword, hybrid and faceted search	PostgreSQL full text for phase 1, OpenSearch from phase 2 if volume requires	Avoid a second cluster until it is justified
Documents	Object storage for files and immutable copies	MinIO, S3-compatible, versioning and object lock enabled	Self-hosted, portable, write-once support for executed copies
Document engineering	Merge, conversion, differences, tracked changes	docxtpl and python-docx for merge, LibreOffice headless for conversion, custom tracked-change writer, Pandoc for extraction	Deterministic and inspectable, no external document service holding our contracts

Layer	Responsibility	Selected technology	Why
Jobs and events	Async work, retries, scheduling, event log	Redis with Celery, plus a durable outbox table in PostgreSQL	Reliability without a message broker to operate in phase 1
Workflow connectors	Mail collection, notifications, calendar writes, glue	Self-hosted n8n, called through signed webhooks	Fast integration work without embedding third-party logic in the core
Identity	Authentication, federation, provisioning	Keycloak federating Microsoft Entra ID and Google Workspace by OIDC or SAML, SCIM provisioning	One role model across two suites, no bespoke authentication code
AI gateway	Model routing, quotas, redaction, logging, failover	LiteLLM proxy in front of Anthropic Claude, Google Gemini, Azure OpenAI and a self-hosted open-weights model on vLLM	Provider neutrality is a configuration change, not a rewrite
AI orchestration	Retrieval, grounding, tool use, evaluation harness	Python orchestration with typed tool contracts and MCP servers for platform data access	Keeps AI access to data inside the same permission model as the application
E-signature	Controlled electronic execution	Self-hosted DocuSeal, with Documenso as the evaluated alternative	Templates, API, audit trail, hosting cost rather than per-seat licensing
Observability	Traces, metrics, logs, AI traces and evaluations	OpenTelemetry, Prometheus, Grafana, Loki, Langfuse for AI tracing and evaluation runs	Standard, self-hostable, and gives per-capability quality visibility
Secrets and keys	Credentials, encryption keys, rotation	HashiCorp Vault, or SOPS with age for the smallest footprint	Keys never in code or environment files in a repository

Layer	Responsibility	Selected technology	Why
Infrastructure	Runtime, deployment, backup	Docker Compose in phase 1, hosted in a self-managed server.	Match operational complexity to team size, keep sovereignty defensible
CI and CD	Build, test, security scan, deploy	GitHub Actions with automated tests, dependency and container scanning, migration checks	Existing engineering practice

10.2 Deployment topology

Environment	Purpose	Data	Access
Development	Feature work, Integration and evaluation runs	Synthetic only	Dev Team
Staging	Release verification, rehearsal candidate migration	De-identified copy, refreshed on demand	Engineering, Legal reviewers
Production	Live legal work	Live records	Role-based, MFA, no direct database access without break-glass approval

Environments are strictly separated. Live legal data never enters development. De-identification for staging removes counterparty names, personal data and monetary values through a documented, tested transformation.

11. Integrations

The organisation runs Microsoft 365 and Google Workspace side by side, so every integration is specified twice or specified through an abstraction. The platform treats both as peers.

11.1 Integration matrix

Need	Microsoft 365	Google Workspace	Platform behaviour
Authentication	Entra ID, OIDC or SAML through Keycloak	Google Workspace SSO through Keycloak	One role model, group-to-role mapping, SCIM provisioning from either
Mail read, approved mailboxes only	Microsoft Graph, Mail.Read on named mailboxes	Gmail API with domain-wide delegation limited to named mailboxes	Poll or subscribe, store message metadata plus body reference, never ingest unnamed mailboxes
Mail send, administrative only	Graph SendMail from the legal mailbox	Gmail API send from the legal mailbox	Templated administrative messages only, human-confirmed, logged
Calendar	Graph Calendars.ReadWrite	Google Calendar API	Obligation and filing deadlines written as events, plus a subscribable feed
Document storage handoff	SharePoint and OneDrive	Google Drive	Optional mirror of final documents into the user's working environment, with the platform copy remaining authoritative
Authoring	Word, desktop and web	Google Docs	Round-trip editing of .docx with tracked changes preserved and the version link maintained
Notification and approval	Teams message with action link	Google Chat message with action link	Same payload, one notification service, per-user channel preference

Need	Microsoft 365	Google Workspace	Platform behaviour
Reporting export	Excel and Power BI dataset	Google Sheets and Looker Studio connector	Read-only dataset endpoints, entity-scoped
E-signature	Self-hosted DocuSeal	Self-hosted DocuSeal	Suite-independent, webhook back to the platform on completion

11.2 Connector governance

Every route by which information can leave the platform or reach another service is registered, owned and reviewed. An unregistered connector is a security incident, not a convenience.

Control	Requirement
Registration	Each connector has a record: purpose, direction, data classes permitted, scopes granted, owner, review date
Least privilege	Scopes are the narrowest that work. Mailbox access is per-mailbox, never tenant-wide. Drive and SharePoint access is per-site or per-folder
Credential handling	Service credentials in Vault, rotated on schedule, never in code, environment files or n8n plaintext
Egress logging	Every outbound call records connector, purpose, record reference, data class and result
Review	Permissions, connectors, retention and AI use are reviewed quarterly, not only at implementation
Failure handling	Connector failure degrades gracefully, queues work, alerts the owner and never silently drops a legal event
Manual fallback	Every connector-dependent workflow has a documented manual path so continuity does not depend on one automation

11.3 n8n usage boundary

n8n is plumbing, not the platform. This boundary is stated explicitly because it is the most likely place for architectural drift.

Permitted in n8n	Not permitted in n8n
Mail polling and message hand-off to the platform API	Legal decision logic, tier assignment or approval routing
Notification fan-out to Teams, Chat, email and SMS	Storing the only copy of any legal record
Calendar event creation and updates	Direct model calls that bypass the AI gateway
File movement between the platform and Drive or SharePoint	Direct database writes
Scheduled triggers for reminders that call platform endpoints	Holding credentials outside the secret manager
Small transformations and format conversion in transit	Sending any substantive external communication

12. Interfaces and APIs

12.1 API principles

- REST over HTTPS with JSON, versioned by path prefix, for example /api/v1. Breaking changes require a new version and a deprecation window of at least 90 days.
- Authentication by OIDC bearer token for users and by client credentials for services. Every service client is registered, scoped and rate-limited.
- Authorisation is enforced in the data layer. An endpoint may not rely on the caller having filtered correctly.
- Every mutating request carries an idempotency key. Retries must not create duplicate matters, documents or signature requests.
- All responses are entity-scoped. A request that would cross entities without explicit permission returns not found rather than forbidden, so record existence is not disclosed.
- Webhooks are signed, replay-protected, retried with exponential backoff and dead-lettered after a defined limit.
- The API is the only write path. The user interface, n8n and any future product client use the same endpoints and the same permission checks.

12.2 Principal endpoint groups

Group	Representative endpoints	Notes
Requests	POST /requests, GET /requests/{id}, POST /requests/{id}/attachments	Mandatory-field validation returns field-level errors
Matters	POST /matters, GET /matters, PATCH /matters/{id}, POST /matters/{id}/transitions, POST /matters/{id}/decisions	Transitions validate the state model in section 8.2
Templates and clauses	GET /templates, POST /templates/{id}/versions, POST /clauses/{id}/versions, POST /versions/{id}/publish	Publication requires the clause-owner role and re-authentication
Generation	POST /documents/generate, GET /documents/{id}, GET /documents/{id}/hash	Deterministic, returns the template and clause versions used

Group	Representative endpoints	Notes
AI capabilities	POST /ai/draft, POST /ai/review, POST /ai/classify, POST /ai/extract, POST /ai/ask	Every response includes sources, confidence, route and an interaction ID for the audit trail
Approvals	POST /approvals/chains, POST /matters/{id}/approvals, POST /approvals/{id}/decision	Decisions bind to a document hash
Signature	POST /signature/requests, POST /signature/requests/{id}/cancel, POST /webhooks/signature	Webhook returns the executed file and audit certificate
Contracts and obligations	POST /contracts, GET /contracts, POST /contracts/{id}/obligations, PATCH /obligations/{id}	Obligation creation from extraction requires a confirmation flag
Counterparties and vendors	POST /counterparties, GET /counterparties?search=, POST /counterparties/{id}/merge	Merge retains prior identifiers as aliases
Assessments	POST /assessments, POST /assessments/{id}/stages/{stage}/complete, POST /assessments/{id}/close	Close requires a residual-risk decision and owner
Reporting	GET /reports/operational, GET /reports/kpi, GET /reports/ai-quality, GET /datasets/{name}	Read-only, entity-scoped, cacheable
Administration	GET and PATCH /config/{area}, GET /audit/events, POST /exports	Configuration changes and exports are audited, bulk export needs second approval

12.3 AI response contract

Every AI endpoint returns the same envelope, so that no consumer can present a model output without its provenance. An output without sources is a failed call, not a low-confidence answer.

Field	Meaning
interaction_id	Identifier joining this call to the audit trail and to the evaluation record

Field	Meaning
capability	The named capability invoked, which determines the tier and the permitted route
output	The suggestion, always labelled as a draft, never as a decision
sources	Ordered references to the retrieved clause, template, matter, contract or decision record supporting each element
unsupported_segments	Any part of the output not attributable to a source, which the interface must mark visibly
confidence	Calibrated score per element where the capability supports it
checks	Results of deterministic validations run on the output, for example defined terms, cross-references, date logic
route	Provider, model, version and parameters used, and the data class the call was permitted for
cost	Tokens and monetary cost, for the AI usage report
requires_human	Always true for substantive outputs, with the role required to confirm

13. AI architecture and governance

AI is a component of this platform, not its identity. The design principle is to automate the predictable first and introduce a model only where interpretation of unstructured material creates genuine value. What follows defines where models are used, how they are constrained, and how their use is proven.

13.1 Four operating categories

Category	Suitable examples	Control position
Deterministic automation	Template population, task creation, approval routing, reminders, calculations, status changes, identifier generation, consistency checks	Preferred wherever the rule can be stated clearly. No generative model involved
Controlled AI assistance	Email classification, metadata and obligation extraction, summarisation, clause comparison, issue spotting, first-draft suggestion, grounded retrieval answers	Approved routes only, scoped data access, grounding required, logging, mandatory human review
Lawyer-led judgment	Interpretation, negotiation strategy, legal advice, risk acceptance, final drafting decisions	A model may assist research or analysis. The lawyer remains responsible and is recorded as such
Restricted or exceptional	Privileged investigations, special-category personal data, credentials, security keys, matters under special secrecy	No general-purpose model. A documented exception plus a specially approved environment or a locally hosted model is required

13.2 Capability register

Every AI use in the platform is a named capability with an owner, a tier ceiling, a permitted data class, a route and a gate. Nothing runs as an unnamed model call.

Capability	Module	Max data class	Human requirement
Inbox classification	M09	Confidential	Legal confirms or corrects before any matter or action is created

Capability	Module	Max data class	Human requirement
Deadline and fact extraction	M09, M08	Confidential	Extracted values are suggestions until confirmed field by field
Administrative acknowledgment draft	M09	Internal	Legal reads and sends. No legal position permitted in the text
Clause retrieval and grounded answer	M10	Confidential	Answer must cite sources. Counsel judges applicability
AI first draft, bespoke agreement	M05	Confidential	Counsel reviews every clause, novel text is flagged and separately confirmed
Deviation detection and severity ranking	M06	Confidential	Counsel reviews all critical and material findings. Legal operations may clear minor findings matching pre-approved fallbacks
Redline suggestion	M06	Confidential	Suggestions are accepted individually and attributed to the accepting counsel
Obligation extraction	M08	Confidential	Legal confirms before obligations become tracked tasks
Template and clause import assistance	M03	Internal	No imported clause is approved without explicit human approval
Management summary generation	M14	Internal	Head of Legal reviews before circulation

13.3 Grounding and retrieval design

- Retrieval is permission-filtered before ranking. The index is partitioned by entity and excludes restricted matters for users without explicit access.
- The corpus is the approved clause and template library, executed agreements, decision records, negotiation outcomes, playbooks and policies. Draft counterparty paper is retrieved only within the matter it belongs to.

- Hybrid retrieval, combining keyword and vector search, with reranking. Chunking is clause-aware rather than fixed-length, so a citation always resolves to a legally meaningful unit.
- Every generated element carries a source reference. Elements that cannot be attributed are returned in `unsupported_segments` and displayed as novel and unapproved.
- Prompts, system instructions and skills are versioned artefacts in the repository, reviewed like code, and referenced by version in every interaction record.
- No client, counterparty or matter content is used to train or fine-tune any model, internal or external. This is an absolute constraint, and the no-training setting is verified per provider and recorded.

13.4 Model routing policy

Data class	Permitted routes	Conditions
Public, Internal	Any approved commercial provider through the gateway	No training use, logging on, standard retention
Confidential	Approved enterprise providers with contractual protections, or the self-hosted model	Zero data retention or shortest available retention, no training use, redaction of unnecessary personal data before the call, full interaction logging
Restricted	Self-hosted open-weights model only, or no model at all	Documented exception approved by Legal, isolated environment, no egress from the platform network

Routing is configuration held in the gateway and mirrored in the platform, and a change to routing is an audited administrative event requiring approval from Legal. Failover between providers is permitted only between routes of equal or stricter class.

13.5 Minimum safeguards before any AI use

Safeguard	Requirement
Approved environment	Only services approved by IT, Legal, Privacy and Security, with enterprise contractual protections in place

Safeguard	Requirement
No training by default	Confirmed per provider in writing, disabled where a setting exists, recorded in the capability register
Data classification	Defined and enforced per service and per capability before the first live call
Least privilege	Mailboxes, sites, libraries, records and connectors scoped to what the capability requires and nothing more
Data minimisation	Redact, pseudonymise or omit anything unnecessary to the task before the call
Retention and residency	Known and documented for prompts, files, logs and outputs, with the shortest workable retention
Connector governance	Every egress route reviewed and registered as in section 11.2
Human verification	A named person reviews accuracy, authority, confidentiality and appropriateness before reliance or external use
Auditability	Service, user, purpose, data class, sources, route, output and final decision recorded
Testing and monitoring	Hallucination, extraction accuracy, bias, leakage, prompt injection and operational failure measured before and after release
Kill switch	Any capability can be disabled instantly, per capability and per agreement type, without deployment
Manual fallback	A documented non-AI path for every AI-assisted workflow

13.6 The trust test

Six controls must hold before any clause is machine-drafted. If one fails, the capability does not run.

Control	Test
Confidentiality	Is it defined what may be shared, and with which tool, for this data class?

Control	Test
Single source of truth	Are only approved playbook clauses presented as house position?
Verification	Are citations, defined terms and cross-references mechanically checked?
Human sign-off	Is it impossible to execute an agreement on model output alone?
Traceability	Is every draft, prompt, source and edit logged and attributable to a person?
Escalation path	Does a named counsel own everything outside the playbook?

13.7 Prompt injection and adversarial input

Counterparty documents and inbound email are untrusted input. This is the most likely route to a serious incident, so it is treated as a security control rather than a quality issue.

- Instructions found inside any ingested document or email are treated as data, never as instructions. System instructions state this explicitly and the orchestration layer strips or neutralises instruction-like content in retrieved context.
- Tool access is allow-listed per capability. No capability may call a tool it does not require, and no model output can widen its own permissions.
- No AI capability may send external communications, alter permissions, publish a clause version, approve an item or trigger a signature request. These actions are unavailable to the model layer by construction, not by instruction.
- Injection attempts are detected, logged, and reported to Legal and IT security, and the affected document is quarantined pending review.

14. Risk tiers and autonomy

Not every contract deserves the same lawyer-hours. The tier determines how much autonomy the platform has and who must sign. Tier is derived by rule, is visible on the matter, and can only be lowered by the Head of Legal with a recorded reason.

14.1 Tier definitions

Tier	Typical scope	Platform autonomy	Human requirement
Tier 1, auto-issue	Standard mutual NDAs and unmodified template agreements below the value threshold	Generated, sent and filed automatically from an approved template with no deviation	Defined Legal review or approved minimal review before issue, plus a monthly quality sample of at least 10 per cent
Tier 2, legal operations cleared	Vendor, consultancy and event contracts on modified templates	AI drafts and flags. Legal operations clears minor deviations that match pre-approved fallbacks	Legal operations confirms every substantive output. Anything not matching a pre-approved fallback escalates to counsel
Tier 3, full legal review	Funder agreements, data-sharing, intellectual property, employment terms, anything with material commercial exposure or sensitive data	AI drafts and flags only	Senior Legal reviews every clause before signature, plus any required functional or management approvals
Tier 4, counsel only	Litigation, mergers and acquisitions, government memoranda, novel, contentious, precedent-setting or highly restricted matters	Substantive automation disabled. Research and retrieval assistance only, on request	Counsel drafts and signs. Any automation requires separate written approval

14.2 Tier assignment rules

Tier is computed from the following inputs, and the highest triggered tier wins. Rules are configuration, not code.

Input	Effect
Agreement type	Base tier per type, set by the Head of Legal
Contract value band	Above the configured threshold raises the tier by at least one level
Counterparty class	Government, funder, regulator or strategic partner raises to Tier 3 or above
Data sensitivity	Personal data raises to Tier 3. Special-category data or privileged content raises to Tier 4
Deviation from template	Any deviation removes Tier 1 eligibility. A critical deviation raises to Tier 3
Novelty	No approved template or clause position for the required term raises to Tier 3 or Tier 4
Precedent effect	Anything likely to set a precedent for future agreements is Tier 4
Term length and exclusivity	Long term, auto-renewal without notice, or exclusivity raises the tier by one level
Cross-border transfer	Transfer of personal data outside Nigeria raises to Tier 3 and triggers a privacy assessment

14.3 Authority to concede

Fallback positions carry an authority level, so a concession is a decision made by someone entitled to make it.

Position	Who may accept	Recorded as
House position	Any authorised user on the matter	Standard, no decision record required
Fallback 1	Counsel	Decision record with reason
Fallback 2	Head of Legal	Decision record with reason and commercial rationale

Position	Who may accept	Recorded as
Fallback 3	Head of Legal plus accountable business owner	Decision record plus explicit residual-risk acceptance
Outside all fallbacks	Head of Legal plus executive sponsor	Decision record, residual-risk acceptance, and a clause library review task

15. Non-functional requirements

15.1 Security requirements

ID	Requirement
LOP-NFR-10	Encryption in transit with TLS 1.3, and at rest with AES-256 for database, object store and backups
LOP-NFR-11	Per-entity, and in phase 4 per-tenant, encryption key separation with documented rotation
LOP-NFR-12	Multi-factor authentication for all users, with step-up authentication for signature, clause publication, restricted access and administration
LOP-NFR-13	Row-level security enforced in the database, so an application defect cannot leak across entity or matter boundaries
LOP-NFR-14	All uploads virus-scanned and content-type verified before storage, with quarantine on failure
LOP-NFR-15	Secrets in a managed secret store, never in source control, images or automation nodes, with rotation on schedule and on staff departure
LOP-NFR-16	Dependency, container and static application security scanning in the pipeline, with build failure on high or critical findings
LOP-NFR-17	Penetration test before the first live matter and annually thereafter, with tracked remediation
LOP-NFR-18	Break-glass access requires two-person approval, is time-boxed, and every action in that session is recorded
LOP-NFR-19	Signed, immutable audit store retained for at least seven years, or the statutory period if longer
LOP-NFR-20	Session timeout, device and location anomaly detection, and enforced logout on role change or deprovisioning

15.2 Privacy and data protection

ID	Requirement
LOP-NFR-21	Compliance with the Nigeria Data Protection Act 2023 and applicable NDPC guidance, with a maintained record of processing activities generated from the platform
LOP-NFR-22	Alignment with the EU General Data Protection Regulation where funder, partner or counterparty agreements require it, including transfer safeguards
LOP-NFR-23	Data minimisation enforced in intake, in extraction and before every model call
LOP-NFR-24	Purpose limitation, with each capability declaring its purpose and the platform blocking use outside it
LOP-NFR-25	Retention schedules per record class, with legal hold, defensible deletion and a deletion certificate
LOP-NFR-26	Data-subject request handling, covering location, extraction, correction and erasure across records, documents and retrieval indexes
LOP-NFR-27	Hosting in Nigeria, or in a region approved by Legal with documented transfer safeguards, with residency recorded per record class
LOP-NFR-28	A DPIA completed and approved for the platform itself before the first live matter, and reassessed at each phase gate

16. Evaluation, testing and quality regime

A capability is not released because it demonstrated well. It is released because it passed a stated gate on a held-out set, and it stays released only while it keeps passing. This section is what converts our AI governance claims into evidence.

16.1 Golden sets

Set	Contents	Size at phase 2	Maintained by
Inbox classification	Real messages from the approved mailbox, de-identified, human-labelled across all categories	Up to 400 messages, all categories represented	Legal operations
Extraction	Emails and executed agreements with human-labelled parties, dates, values, deliverables and obligations	Up to 200 documents	Legal operations
Clause retrieval	Questions with known correct clause answers from the approved library	Up to 150 questions	Counsel
Drafting	Briefs with counsel-reviewed reference drafts and a defect rubric	Up to 60 briefs across five agreement types	Counsel
Deviation detection	Counterparty drafts with human-labelled deviations and severities	Up to 80 documents including edge cases and absent clauses	Counsel
Adversarial	Prompt injection, exfiltration attempts, contradictory instructions, poisoned documents, out-of-scope requests	Up to 100 cases, growing with every real incident	Legal operations

Golden sets are held out of any prompt or configuration development. Anyone tuning a capability may see aggregate results, not individual held-out items. Every correction made by Legal in normal use is a candidate for the sets, and the sets are refreshed monthly so they track real work rather than launch conditions.

16.2 Test levels

Level	Scope	Automation	Gate
Unit	Business rules, tier computation, validation, identifier generation, state transitions	Full, in the pipeline	All pass, coverage on lifecycle and permission logic above 85 per cent
Integration	API contracts, database permissions, connectors, webhooks, generation determinism	Full, in the pipeline	All pass, including a determinism test that regenerates and compares hashes
Permission and isolation	Entity separation, restricted matters, retrieval filtering, tenant isolation	Automated adversarial suite run every build	Zero leakage. Any single failure blocks release
AI capability evaluation	Golden sets against the gates in section 4.2	Automated harness, run on change and monthly	Capability meets its gate, otherwise it stays disabled
Adversarial and security	Injection, exfiltration, jailbreak, malformed documents, oversized inputs	Automated suite plus quarterly manual red team	Zero successful exfiltration or unauthorised action
User acceptance	Real work by Legal, Legal operations and requesters in staging	Manual, scripted scenarios	Sign-off by the Legal product owner against acceptance criteria
Operational readiness	Backup and restore, failover, capability disablement, runbook walkthrough	Manual, before each phase go-live	Successful restore test and a rehearsed kill switch

16.3 Monitoring in production

- Every AI interaction is traced with capability, route, sources, latency, cost and the human decision that followed it.

- Human correction rate per capability is the primary live quality signal. A sustained rise of more than 20 per cent over a fortnight triggers investigation and may trigger automatic disablement.
- Drift monitoring compares live input distribution to the golden set. Significant drift triggers a golden set refresh rather than a silent tolerance of degradation.
- Cost and latency per capability are reported weekly, so a route change is a visible decision with a visible effect.
- A monthly quality sample of at least 10 per cent of Tier 1 auto-issued documents is reviewed by Legal, and results are recorded against the KPI set.
- Every incident, including near-misses, produces a written record with cause, containment, correction and a golden set addition where applicable.

16.4 Traps that make AI legal work fail

These are recorded as explicit anti-requirements, because each has a known failure mode in comparable programmes.

Trap	How this platform prevents it
Automating an ambiguous process	The lifecycle, tiers, owners and approval chains are defined before automation. Automation of an undefined process is refused at design review
AI drafting against uncontrolled templates	M03 must be published and approved before M04 and M05 are enabled for any agreement type
Ungrounded output that reads well	Sources are mandatory, unsupported segments are surfaced and marked, and deterministic checks run before presentation
Silent scope creep in autonomy	Tier ceilings are configuration, changes are audited, and no capability can raise its own tier
Shadow tools filling the gaps	The portal must be faster than the habit it replaces, and unregistered connectors are treated as incidents
Metrics without a baseline	Baselines are captured in weeks 1 and 2, and no target is accepted as met without one

Trap	How this platform prevents it
Governance as a document rather than a control	Every control in section 13 has a corresponding test in section 16.2 and a report in M14
One person holding the knowledge	Decision records are mandatory for concessions, and institutional memory is built from normal work rather than a separate task

17. Roadmap and delivery plan

17.1 Phase 0, foundation, weeks 1 to 2

Work	Output
Confirm requirements, data classifications, owners, first two templates and baseline measures	Approved build design and control checklist
Stand up environments, identity federation to both suites, repository, pipeline, secret management	Working development and test environments with single sign-on
Complete the platform DPIA and the security design review	Approved assessment and a remediation list before any live data
Capture baselines for every KPI in section 4.1	Recorded baseline with the measurement method for each metric

17.2 Phase 1, internal pilot, weeks 3 to 4

Weeks	Work	Output
4 to 6	M15 access control and audit, M01 intake, M02 matter records and triage, counterparty records	A working front door and matter register in test
7 to 9	M03 core library for two agreement types, M04 deterministic generation, M07 approvals and e-signature	End-to-end generation to signature for two templates
10 to 11	M08 archive and obligation tasks, M14 core dashboards and the weekly update generator	Post-signature tracking and live reporting
12	Tier 1 auto-issue for standard NDAs, Tier 2 assisted classification and extraction on the approved mailbox in shadow mode	First AI capabilities running with mandatory review

Weeks	Work	Output
13	User testing, privacy and security review, training, controlled live use and measurement	Go-live decision and a phase 2 business case with measured results

Shadow mode means the capability runs and records its output for comparison but its suggestions are not shown as recommendations to users. A capability leaves shadow mode only after meeting its gate in section 4.2.

17.3 Phase 2, complete internal platform, weeks 5 to 6

Work	Output
M03 full clause library with fallbacks and authority levels, plus playbooks per agreement type	House position captured as a controlled asset
M05 AI drafting for bespoke agreements, with provenance and consistency checking	First-draft time for bespoke work materially reduced
M06 review, deviation flagging and tracked-change redlining	Counterparty paper reviewed against the playbook with ranked findings
M09 inbox intelligence out of shadow mode, M10 institutional memory retrieval	Implied work surfaced, and a searchable memory in daily use
M11 DPIA and AI assessment workflows, M12 compliance calendar, M13 vendor governance	Privacy and compliance operating as workflows with evidence
Expansion to further templates and both entities at full volume	Platform is the normal way legal work happens

17.4 Sequencing constraints

Constraint	Consequence if ignored
No generative drafting before the clause library is published and approved	Ungrounded drafts become precedent, and the audit trail points to nothing authoritative
No inbox intelligence before matter triage exists	Recommendations have nowhere to go and Legal loses trust in the first week
No obligation extraction before the executed archive is authoritative	Obligations attach to drafts and diverge from what was signed
No external tenant before isolation is independently verified	A single leakage event ends the product proposition
No new AI capability while an existing one is below its gate	Quality debt compounds and the governance claim becomes untrue

18. Risks and mitigations

Risk	Impact	Mitigation
Over-automation of legal judgment	High	Rules-based workflow separated from AI assistance, tier ceilings, mandatory human confirmation, and actions such as sending, approving and signing unavailable to the model layer by construction
Sensitive data leakage through a model or connector	High	Classification, least privilege, routing policy by data class, redaction before calls, egress logging, connector registration, automated isolation tests every build
Prompt injection from counterparty documents or email	Medium	Untrusted-input handling, allow-listed tools, no autonomous actions, adversarial golden set, quarterly red team, quarantine on detection
Ungrounded output relied upon	High	Mandatory sources, unsupported segments surfaced, deterministic consistency checks, drafting gate on unsupported statement rate
Poor adoption, work continues by email	High	Intake faster than the habit it replaces, requests raisable from Teams and Google Chat, no duplicate entry, training by role, and portal share of intake tracked as a KPI
Clause library never actually gets built	High	Library is on the critical path with a named owner and a phase 1 deliverable. Drafting features stay disabled until it is published
Key-person dependency in a small build team	Medium	Standard open-source stack, infrastructure as code, documented runbooks, no bespoke identity or storage layer, pair coverage on the AI layer
Scope expansion	Medium	Written backlog, non-goals in section 3.3, change requests only at phase boundaries unless the sponsor accepts a trade
Bad data migrated from the existing register	Medium	Migrate only active records, define mandatory fields, clean before load, reconcile counts, keep the workbook read-only as a historical reference

Risk	Impact	Mitigation
Two suites double the integration cost	Medium	One abstraction per capability with two adapters, and no feature depending on a single suite
Build effort underestimated because it is bespoke	Medium	Modular monolith rather than microservices, adopt identity, signature and connectors rather than building them, phase gates with measured evidence
Regulatory change in data protection or AI	Medium	Assessment records designed for reuse, review dates, provider terms and residency recorded per record class